The Trials & Tribulations of the MCDC

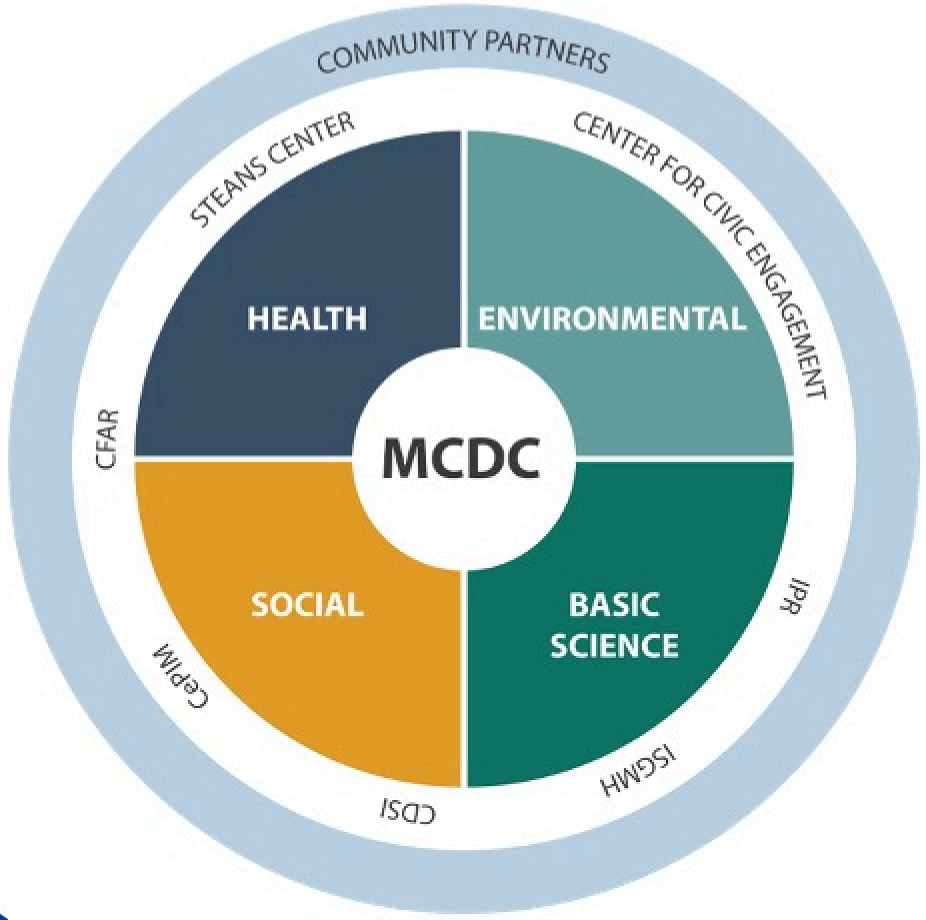
Phil Yates
Joint Statistical Meetings, Boston, MA
August 5, 2026

DePaul University

Metropolitan Chicago Data-Science Corps (MCDC)

- Established and funded by an NSF grant (Award number: 2123401) from September 2021 to August 2024
- The main pillars of the MCDC were to advance the development of data acumen via the combination of **university expertise and curriculum**, **community partners**, and **student learning and engagement**
- Participating institutions: Northwestern, Northeastern Illinois, DePaul, University of Illinois Urbana-Champaign School of Information Sciences, Chicago State

Linking to Community Partners



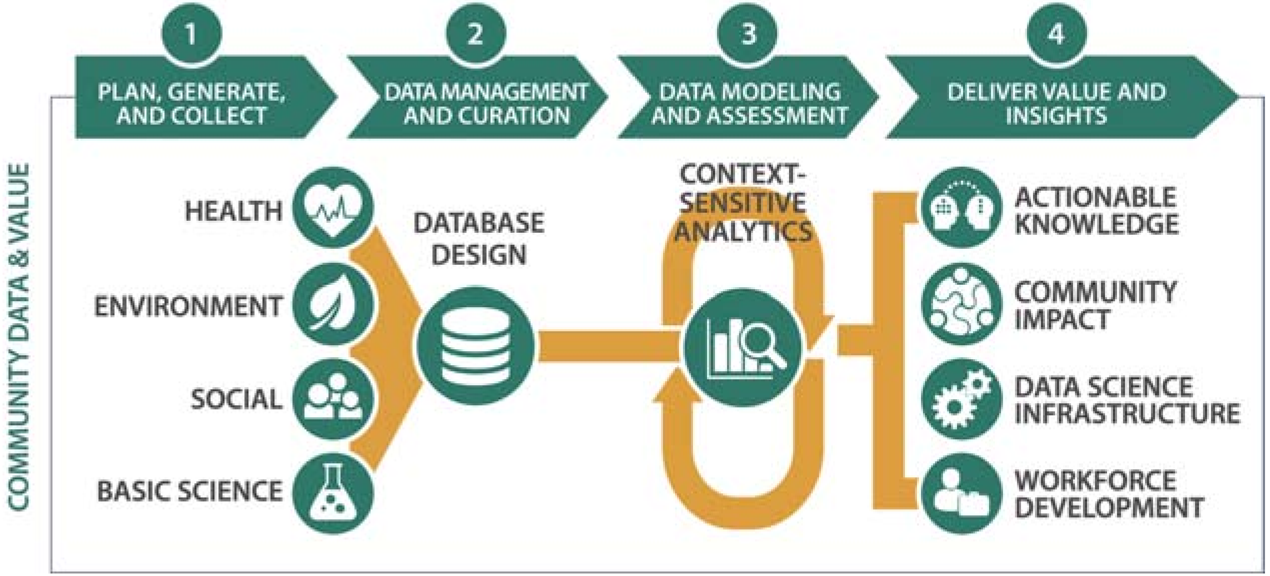
Scientists and students in the MCDC "inner circle" can connect to community partners in the "outer circle" via existing collaborative centers.

DePaul University: Steans Center, Center for Data Science (CDSI)

Northwestern University: Center for Civic Engagement, Institute for Policy Research (IPR), Institute for Sexual and Gender Minority Health and Wellbeing (ISGMH), Center for Prevention Implementation Methodology, Third Coast Center for AIDS Research (CFAR)

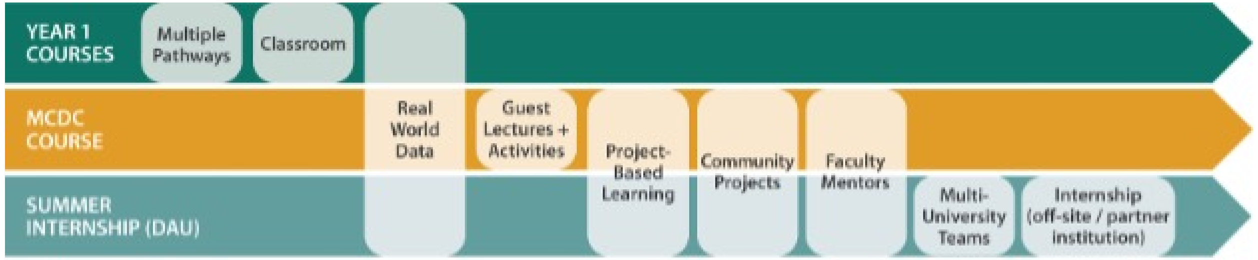
Linking to Community Partners

Students participating in the MCDC learn about all stages of data refinery



Curriculum

Integration of MCDC into a data science curriculum occurs at three key points:



1. **Year 1 - Pathway courses:** Existing data science and data-driven domain courses that use real-world data and together present multiple pathways of entry into MCDC
2. **Year 2 - Practicum courses:** A project-based MCDC course in which student groups work with community partners under the guidance of faculty mentors
3. **Summer term after Year 2:** An optional, selective MCDC Data Science Application for Undergraduates (DAU) in which cross-institutional student groups each work with an engaged community partner on a specific project, embedded with the community partner (internship) or with a faculty mentor at one of the implementation institutions

Year 1 - Pathway Courses

At DePaul University, there were four different three-course sequences of pathway courses.

Data Science (College of Computing and Digital Media): Data Analysis (IT 223); Data Analysis and Regression (DSC 323) **or** Intro to Image Processing (CSC 381); Foundation of Data Science (DSC 441) **or** Applied Image Analysis (CSC 382)

Data Science (Mathematical Sciences): Statistical Methods Using SAS (MAT 341); Applied Statistical Methods (MAT 348); Applied Probability (MAT 349)

Geography and GIS: Digital Mapping (GEO 141); Community GIS (GEO 242); Statistical Data Analysis for GIS (GEO 391) **or** Programming in Python for GIS (GEO 345) **or** Spatial Analysis (GEO 344) **or** Spatial Data Science (GEO 348)

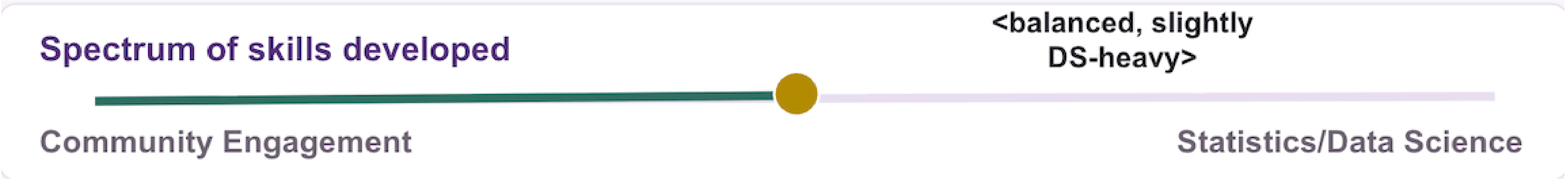
Environmental Science and Studies: Applied Ecology (ENV 250); Environmental Data Analysis (ENV 260); Advanced Environmental Data Analysis with R (ENV 359)

Year 2 - Practicum Courses

ENV 359 (Advanced Environmental Data Analysis with R) and GEO 242 (Community GIS)

Typical format of these courses:

- **Weeks 1 to 4:** introduction to the course and ins and outs of working with community partners, review of various data science and GIS topics
- **Weeks 5 to 9:** working on the projects with and for the community partners, present more advanced data analysis topics as needed when they arise from the projects' work
- **Week 10:** present the final projects
- **Finals week:** turn in the final project



Summer Term After Year 2 - DAU

Example from Summer 2023:

- A team of two DePaul students and one Northwestern student worked on this project presented by the CivicLab on Tax Increment Financing (TIF) Districts in Chicago
- **Question:** How do the community activists associated with the CivicLab and the TIF Illumination Project summarizes information from each of Chicago's 131 (as of summer 2023) TIF districts?
    - Each TIF district has a 35- to 40-page PDF associated with it.
- **Goal:** Develop an efficient way to scrape data from these individual PDF's and effectively summarize and visualize the data for the community activists.

Summer Term After Year 2 - DAU

GitHub: https://github.com/philipayates/chicago2022TIF

Shiny App: https://philipayates.shinyapps.io/apps/

MCDC Evaluations

Practicum Survey of Students (Highlights):

- Students liked tackling real world problems; teamwork
- Students felt like they still needed communication between team and community partners; better technical skills
- Obstacles included time, communication and working remote, and data issues

Practicum Survey of Community Partners (Highlights):

- *Benefits:* The final deck of slides provided by students can be used in grant writing and the community partners' advocacy work; building new collaborations in the Chicago region
- *Obstacles:* Project period was a little too short; students did not have time to really dig into the data

DAU Internship Survey (Highlights):

- Students liked having an impact on the world by using data analytics; working with people from other institutions with different backgrounds; learning from and collaborating with each other
- Students gained insight into a real-life scenario in the workplace; presenting for clients; working with new people